

From DevOps to GovOps:

The Application of Engineering Principles for the Continuous Optimization of Regulatory Environments

Disclaimer: *This is an illustrative sample. This document is meant to introduce the concept of regulatory optimization and illustrate how legal, regulatory rules, and administrative workflows can be represented, modernized, and optimized. It is not a mechanism for deregulation. It is agnostic to the number of regulations a state or jurisdiction writes, or the number of protections a legislature attempts to confer. Its purpose is to optimize both that protection and ease of doing business – by giving law makers visibility into how this affects permitting timelines in their area, how their proposals compare to similar proposals in the country, and how to use that information to implement the most efficient regulatory frameworks possible. This is not a finished case study demonstrating that the architecture has worked, nor is it a detailed spec document detailing explicitly how to implement this proposal. Its purpose is to introduce a concept and invite further discussion.*

Executive Overview

We have different words for it – Vetocracy¹ – Kludgocracy² – Everything Bagel governance.³ But the failure is familiar: Governments increasingly fail at executing on their own goals even when expertise, funding, and clear democratic mandates are all present.⁴ The problem is a cumulative one. The state has accumulated layers of statutes, regulations, court decisions, and agency procedures as a defense for past abuse, but the result is a regulatory system that no institution can fully read, parse or maintain.

While each of these reforms made sense in isolation, their aggregate effect has brought operational throughput to a crawl⁵, and costs have exploded relative to peer countries.⁶ The political debate has often defaulted to a dichotomy between maintaining the opaque and systemic ‘kludge’ as it was implemented, or broadly removing safeguards writ large without regard for which ones serve a legitimate public purpose, and which ones just increase the bureaucratic lift required to accomplish a task.

This proposal supplies the missing maintenance layer for these systems. It links the legal text of a regulation with a direct operational schema to observe how that text is translated to agency procedures. It enables direct comparison of both protective indices (e.g. clean air, clean water), operational throughput, and legal text across state and local jurisdictions. It proposes a regulatory sandbox structure that compares a proposed change with matched controls. And

¹ Fukuyama, Francis. Political Order and Political Decay: From the Industrial Revolution to the Globalization of Democracy. New York: Farrar, Straus and Giroux, 2014. pg. 514–517.

² Teles, Steven M. Kludgeocracy: The American Way of Policy. Washington, D.C.: New America Foundation, 2012. pg. 5–6.

³ Klein, Ezra, and Derek Thompson. Abundance. Avid Reader Press, 2025. pg. 113.

⁴ Klein, Ezra, and Derek Thompson. Abundance. Avid Reader Press, 2025. pg. 71–78.

⁵ Teles, S. M. (2013). “Kludgeocracy in America.” National Affairs, 17, 97–114.

⁶ Klein, Ezra, and Derek Thompson. Abundance. Avid Reader Press, 2025. pg. 77.

finally, it maintains it all in a machine readable and version controlled environment that enables line-by-line traceback to which lawmakers proposed which proposal, with merge packets detailing the exact changes in throughput and protective indices.

None of this involves fundamentally new or unproven technology. The proposal combines capabilities already used across the RegTech industry, regulatory information management, software engineering, and data science. Platforms such as Regology⁷ and CUBE⁸ already structure and compare regulatory text, track compliance and risk, provide Retrieval Augmented Generation (RAG) style AI search services, and help companies map regulatory text to operational workflows. Regulatory information management and PUC e-filing systems provide systems of record for documents, cases, and administrative workflows, while Git-style version control and RAG provide mature models for managing changes and querying complex information. What this proposal adds is a standardized architecture connecting these capabilities at the clause level of each regulation – linking legal text to the workflows and dependencies it creates, providing a common layer and enabling the software ecosystem that has been built on top of these workflows.

1. The Implementation Crisis

At their core, regulations are a set of instructions – building a house requires permits from Agency X – building a factory requires approval from Agency Y – wastewater cannot be disposed without meeting standard Z. This is the critical software that governs how things get done in the real world.

But unlike modern data and software systems which enable the full spectrum of ‘Create’, ‘Read’, ‘Update’, and ‘Delete’ functionality, most state legislatures are only operating at a ‘Create’ and ‘Delete’ level of functionality. Similar issues have been solved in the private sector, as Big Tech companies have utilized mono-repositories to govern systems that serve billions of active users and are continuously worked on by thousands of engineers simultaneously. They accomplish this complexity management through systems that enable structured representations, dependency mapping, observability, version control, controlled experimentation, and continuous improvement.

This proposal adapts those disciplines to government administration with the objective being: make systems observable, testable, auditable, and revisable. As a design constraint disclaimer – The legal text remains the law. The dual schema models are traceable workflow projections, but changing the underlying legal text still requires approval from lawmakers. Furthermore, a sandbox produces evidence rather than law, and deployment still requires approval from an applicable authority.

2. Dual-Schema Regulatory Architecture

⁷ Regology, “Regulatory Intelligence Platform”, <https://www.regology.com/>.

⁸ CUBE, “RegPlatform”, <https://www.cube.global/products/regplatform>.

This proposal represents every regulatory object through two linked representations. The legal schema contains the legal text of the legal object (be that regulation, order, or judicial interpretation) and is pre-pended with structured metadata that codebase management software can use to categorize the document by jurisdiction, relevant agency, effective date, scope, amendments, dependencies, etc.

The workflow schema connects to the first schema via foreign key and models the implementation details it produces (be that stages, documentation, agency reviews, inspections, permits, appeals, etc.). Persistent identifiers link each clause of legal text to its consequential workflow implementation.

This connection is maintained by representing the legal text as a machine-readable (markdown) document. Each legal document is stored in a repository that allows broader analysis, and each document contains a YAML header that allows software to recognize, organize, and understand it.

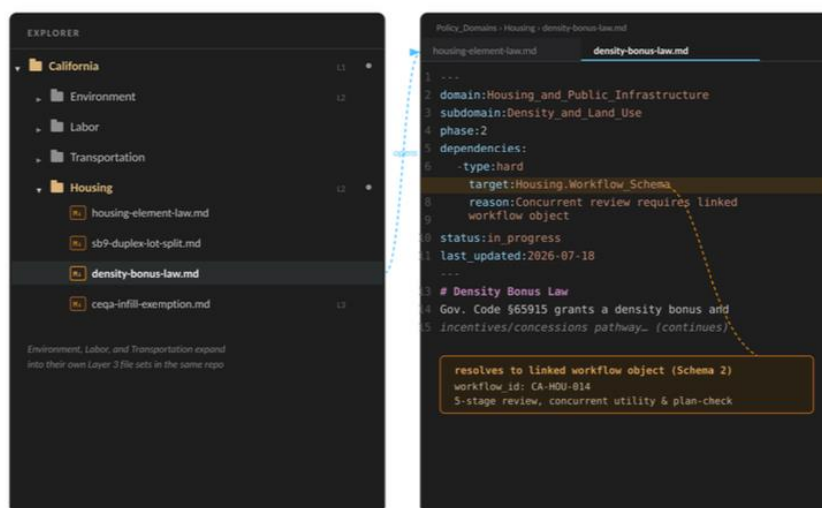


Figure 1 – Repository Organization: Each policy document is stored in a folder directory regarding its jurisdiction and domain. Users would be able to navigate the directory to quickly find the legal text of Colorado’s environmental policy, for example.

```
---
domain: [Domain_Name]
subdomain: [Subdomain_Name]
phase: [0-6]
dependencies:
  - type: [hard|soft|optional]
    target: [Domain.Subdomain]
    reason: [Brief explanation]
status: [draft| in_progress | review | complete ]
last_updated: [YYYY-MM-DD]
---
```

Figure 2 – YAML metadata example.

The YAML frontmatter on the head of each document enables database-style operations to quickly compare the legal text between jurisdictions like California, Alabama, and Colorado. It also enables the clauses of each document to be explicitly linked to the operational workflows they create.

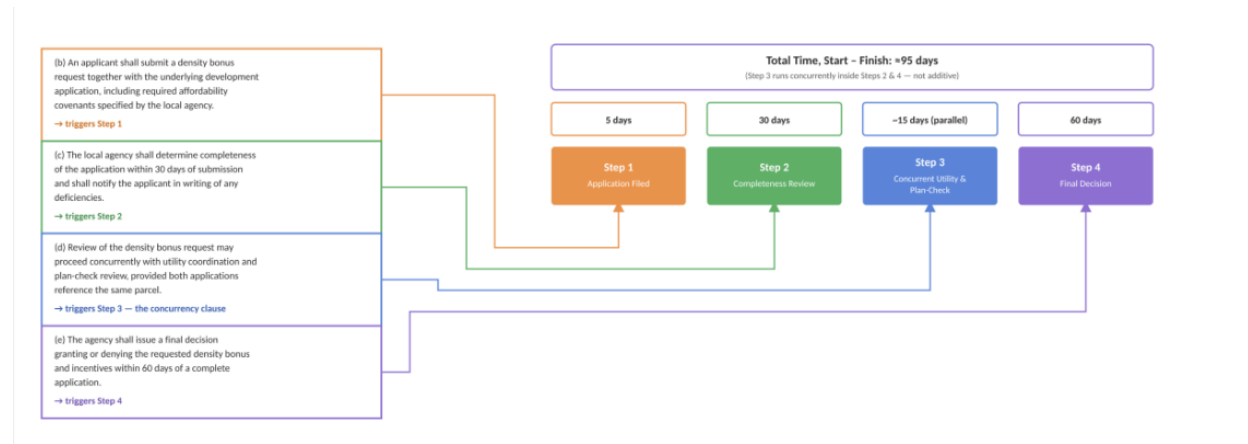


Figure 3 - Dual Schema Representation: Figure represents how YAML front matter can be used to explicitly link each clause of legal text to the workflow stage that it creates.

When implemented on a national scale, this standard allows local governments to maintain their own repositories, contract with their own vendors (or build their own systems on top) while maintaining a common architecture and APIs enabling each government to compare regulatory landscapes, operational throughput, and version histories across jurisdictions. This step also enables higher-level comparative functions.

3. Workflow Intelligence and Regulatory Comparison

The first of the comparative functions that this proposal enables is the direct comparison of operational systems that jurisdictional legal text produces.



Figure 4 – Regulatory Workflow Comparison: This figure represents the temporal regulatory workflow comparison of state-level jurisdictions for an example infrastructure project.

Imagine the regulatory approvals needed to build a solar farm – the permits (and their applicable timelines) needed in order to accomplish the work. In this example, California represents the baseline. Colorado preserves an identical sequence of regulatory steps while completing each stage more efficiently, thus completing the process with 60 days to spare. Connecticut, despite taking longer than Colorado on each individual step, executes steps 1 and 2 concurrently and thus shaves 20 days off of the regulatory approval time needed in Colorado (despite each step taking longer in isolation). Texas follows a fundamentally different strategy by omitting intermediate review stages entirely. This turns their regulatory regime into a two-step process that comprises what would be step 1 and step 4 in any other regulatory regime measured. This represents the ‘deregulation’ approach, which shaves the regulatory approval process down to 45 days – the shortest on this list – but also comes at the cost of a lower protective index.

This illustrates the type of analysis that this proposal can produce. It distinguishes between faster execution of the same protections, concurrent sequencing of independent reviews, removal of duplicated steps, and the removal of legitimate protection. In this instance, California and Colorado can learn from Connecticut’s concurrent sequencing of steps 1 and 2. Likewise, Connecticut and California can learn from Colorado’s executional efficiency, which allows it to execute the same steps in less time. Each state can learn from Texas if the removal of intermediary steps does not translate to a substantive loss in protections.

This turns the classic “more versus less regulation” debate into a diagnosis of stages, dependencies, handoffs, durations, and outcomes, all the while giving lawmakers traceability into the applicable letter of the law. It also enables the type of analysis illustrated in Figure 5.

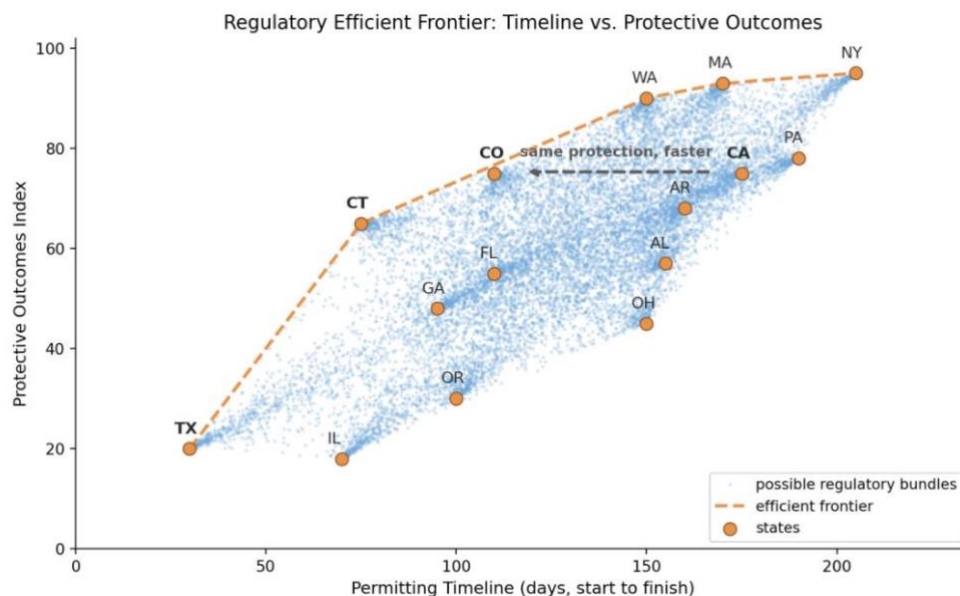


Figure 5 - Efficient Frontier Concept for Regulatory Environments: The efficient frontier is a concept borrowed from finance that identifies the asset portfolio yielding the lowest risk for a targeted return. Here, it is applied here to regulatory drafting instead. By mapping this frontier, we can determine the most efficient combination of regulatory clauses required to achieve a specific level of protection for a targeted permitting timeline.

An Efficient Frontier concept – borrowed from the finance industry – can also be applied to government regulations: jurisdictions plotted by median permitting timeline against an aggregate Protective Outcomes Index lay bare the cost of traditional ‘de-regulation.’ This protective outcomes index – although not specified here – is calculated from domain-specific, already-existing empirical measures (OSHA-style incident rates for jobsite safety, water quality sampling data for clean water). Each regulation (or regulatory category) would have its own index to measure against permitting and regulatory throughput. States that seek to increase operational throughput at the expense of protection increase their risk relative to other jurisdictions. Cutting and recombining the individual clauses of legal text across states illustrates the permitting throughput vs. protective outcomes that would result at other points along the frontier — the highest operational throughput achievable for a given level of risk.

Importantly, while this gives lawmakers legibility into how they can improve the efficiency of their regulatory regimes, it does not by itself give states a way to test new regulatory methods in ways that are safe, bounded, and produce the data they need to make informed decisions. This is detailed in Part 4.

4. Experimental Governance: The Scientific Methodology, Applied to Government Systems

Despite the legibility offered by the systems above, experiments in the real world have established that ‘theory’ is not entirely consistent with real life. Therefore, lawmakers need a structured and easy-to-use methodology for testing changes to their state’s regulatory regime in a way that is (as aforementioned) safe, bounded, and automatically produces the data they need to make informed decisions about scale or sunset.

To respond to this question, the Regulatory Modernization Sandbox (RMS) is proposed. The RMS is a legally authorized ‘branch’ of the regulatory system, in the same sense that codebase changes are ‘branched’ and tested in a bounded and isolated environment. Each sandbox can either be administered digitally within the software system itself or (optionally) directly administered by a Regulatory Modernization Corps (RMC) deployment – government personnel that embed alongside the agencies responsible for executing the workflow under study, or to the site of the regulatory change (alongside local stakeholders) to directly observe how changes affect operations on the ground.

This setup ensures that friction points that occur during the process are identified directly and dealt with up front – instead of waiting years for feedback to filter through complaints, hearings, or elections themselves.

Site Selection and Experimental Design

Sandbox jurisdictions are selected for causal identification rather than convenience. As an illustrative walkthrough, consider a Colorado housing-permitting design. Suppose a change to housing regulations is proposed, and lawmakers want to discern how this proposal changes the amount of time it takes to get the permits needed to build a housing development.

Candidate sandbox jurisdictions are selected by comparing features applicable to median permitting durations and permitting volumes in Colorado jurisdictions. From there, sites are selected according to how well they match up to synthetic control jurisdictions, as identified through Standardized Euclidean Distance, propensity scores, Mahalanobis distance, or synthetic control weights – the specific matching will be chosen according to the best fit for that regulatory domain. As an example:

Jurisdiction	Population	Median Household Income	Population Density	Owner-Occupied Housing	Median Age	Permitting Volume	Median Permitting Duration
Pueblo	112,000	\$54,000	3,200	61%	39	<i>Illustrative</i>	<i>Illustrative</i>
Greeley	111,000	\$60,000	3,700	63%	36	<i>Illustrative</i>	<i>Illustrative</i>
Longmont	101,000	\$82,000	3,900	58%	37	<i>Illustrative</i>	<i>Illustrative</i>
Loveland	82,000	\$76,000	3,400	64%	41	<i>Illustrative</i>	<i>Illustrative</i>

Figure 6 - The values above are illustrative and are meant to exemplify how candidate sites can be selected using nearest-neighbors matching from candidate metrics. Each row represents a potential treatment or control jurisdiction. Each column represents a standardized demographic, economic, housing-adjacent, or administrative characteristic used in the matching procedure. An applicable model would include a more robust feature space (featuring target-variable-relevant cleaning, feature selection, etc.).

From there, applicable treatment-control jurisdictional pairs are selected according to their eligibility, using matching algorithms on the feature matrix to find pairs of jurisdictions that are ‘close enough’ to compare against each other with statistical power.

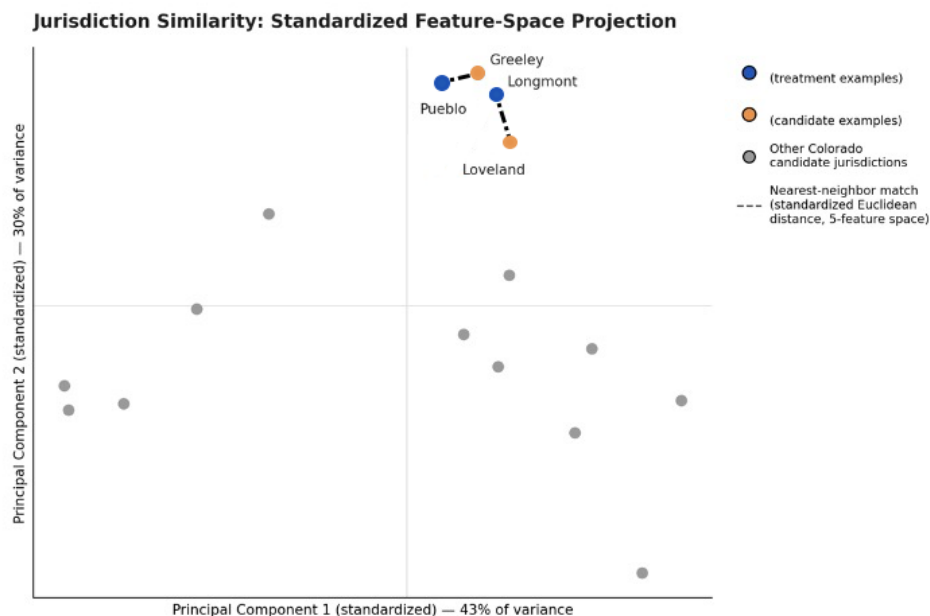


Figure 6b - The values above are illustrative and are meant to exemplify how candidate sites can be selected using nearest-neighbors matching from candidate metrics. Each row represents a potential treatment or control jurisdiction. Each column represents a standardized demographic, economic, housing-adjacent, or administrative characteristic used in the matching procedure. An applicable model would include a more robust feature space (featuring target-variable-relevant cleaning, feature selection, etc.).

Notably, this matching criteria does not include geographic closeness – the final feature space could be north of dozens of applicable dimensions. Once applicable jurisdictions are selected, they undergo a time-bounded sandbox treatment of potential regulatory changes.

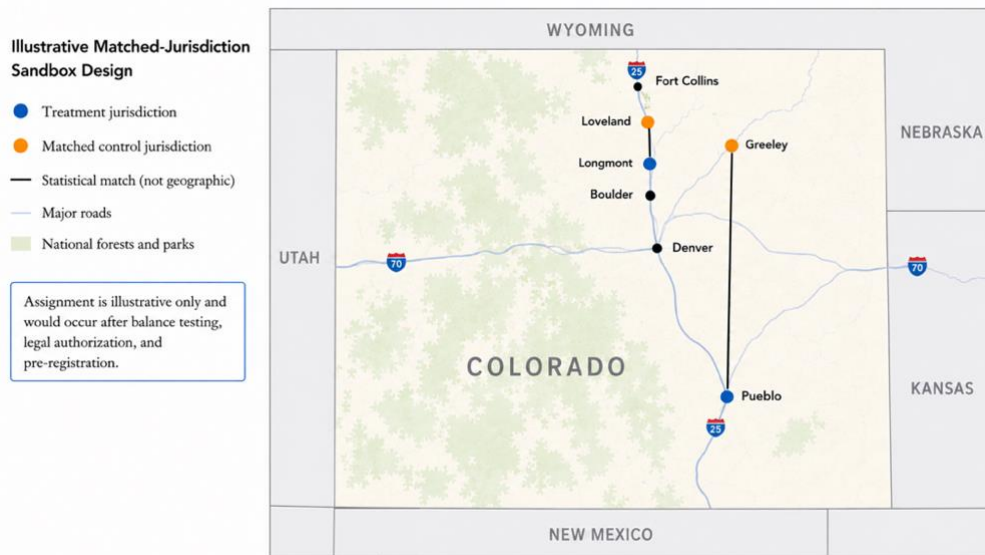


Figure 6c – Geographical Distribution: The geographic distribution of cities designated as ‘treatment’ or ‘control’ are not necessarily geographically correlated. Instead, pairings are chosen via characteristics such as permitting throughput, population, construction characteristics, etc.

Evaluation

Once the treatment-control pairs are selected, a Regulatory Modernization Packet (RMP) is prepared to test changes to the regulatory environment and how these changes affect both the median permitting duration as well as the relative protective index. This setup allows lawmakers to test regulatory changes in a bounded manner while minimizing the risk of immediate statewide rollout.

For example, consider a Difference-in-Differences design comparing the median housing permitting time for one matched jurisdictional pair. The line in the middle (designated as T_0) represents the time point in which a regulatory change is made. In this illustration, the ‘treatment’ jurisdiction (Longmont) experiences a reduction in median permitting time duration relative to its matched control (Loveland).

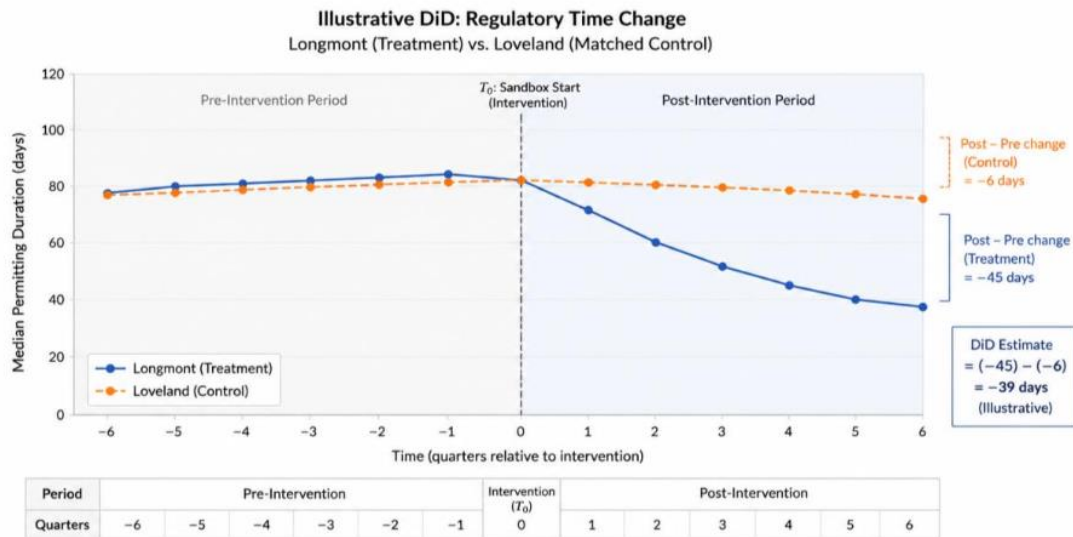


Figure 7 – Regulatory Change Illustration: Once a regulatory change is tested, causal analysis will take place comparing the ‘treatment’ jurisdiction (in this example, Longmont) with the matched control jurisdiction (in this case Loveland).

To increase the statistical rigor, outcomes can also be evaluated using regression discontinuity at jurisdictional boundaries. This is an important step, as geographically separated areas might use different builders, staff, contractors, material suppliers, and so on. The addition of a regression discontinuity at a jurisdictional edge allows us to more effectively isolate the effect of the regulatory change, as housing development projects close to Longmont will naturally use similar contractors, staff, builders, etc.

While multiple experimental designs running concurrently may seem like an over-engineered way to evaluate a regulatory change, it is important to remember that the real world is not a laboratory setting where everything is controlled. Because of this, multiple designs are needed in order to isolate the change in permitting time is actually due to the regulatory proposal and not due to the clearance of a local permitting queue, the efficiency of a local contractor, or so on. Incorporating multiple data streams to run these tests is a mature, industry-standard workflow in software environments – and can be fully automated. If fully implemented, this proposal would not require expensive personnel deployments to collect the necessary data. That data is an emergent property of the dual schema architecture proposed in part 3, which could be pulled automatically to allow lawmakers to see how that effects permitting throughput from multiple angles.

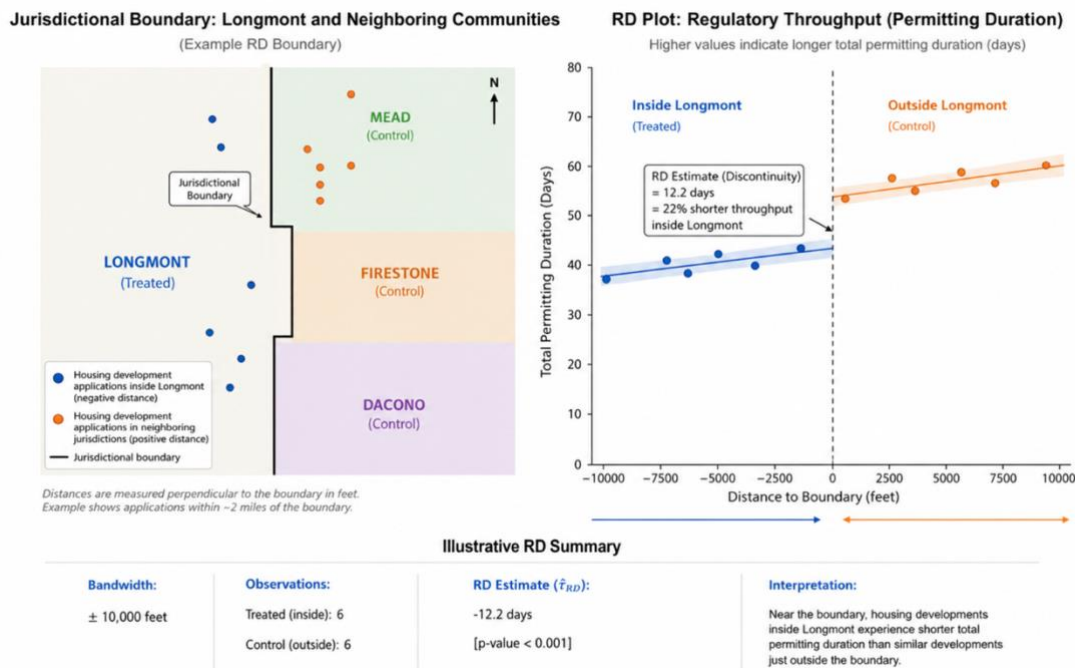


Figure 8 – Regression Discontinuity Illustration: Once a city is designated as a ‘treatment’ jurisdiction for a regulatory change, data is automatically pulled for building projects around its periphery and used to rule out changes in housing construction speed are due to permitting throughput and not the efficiency of any particular builder or supplier.

Independent Review and the Dual Key

Importantly, the government cannot be allowed to design, measure, and certify its own experiments – otherwise we are recreating the same ‘grade your own work’ failure that this setup is meant to avoid. As such, the methods above are combined with a dual-key evidence system built to replicate the peer review structure in scientific fields.

As an overview, this would work by an official evaluating body certifying: baseline conditions, data collection methods, and statistical methods. A separate Independent Research Review panel – composed of accredited universities, nonpartisan research institutions, or other public policy labs – evaluates the site selection, experimental design, data collection, statistical validity, reproducibility, and interpretation of the findings.

Disclaimer: While these methods are good statistical approximations of a controlled experiment, they are not equivalent to laboratory control. Statistical techniques can construct a counterfactual, and strengthen causal evidence, but they cannot eliminate all potential confounding variables or produce irrefutable conclusions. Those limitations do not constitute an argument against evaluation in this manner. Even imperfect causal measurement is a substantial improvement over the current status quo, in which policies are often implemented without a rigorous mechanism for determining whether they can achieve their intended outcomes.

5. From Sandbox to Statute

Upon the completion of a successful pilot, the data and pilot results are automatically incorporated into a Regulatory Modernization Packet (RMP) which contains the amendment to the regulatory environment – containing a ‘legal diff’ of the exact language change to statute, a ‘workflow diff’ of the dual-key certified workflow change, and any applicable notes or evidence.

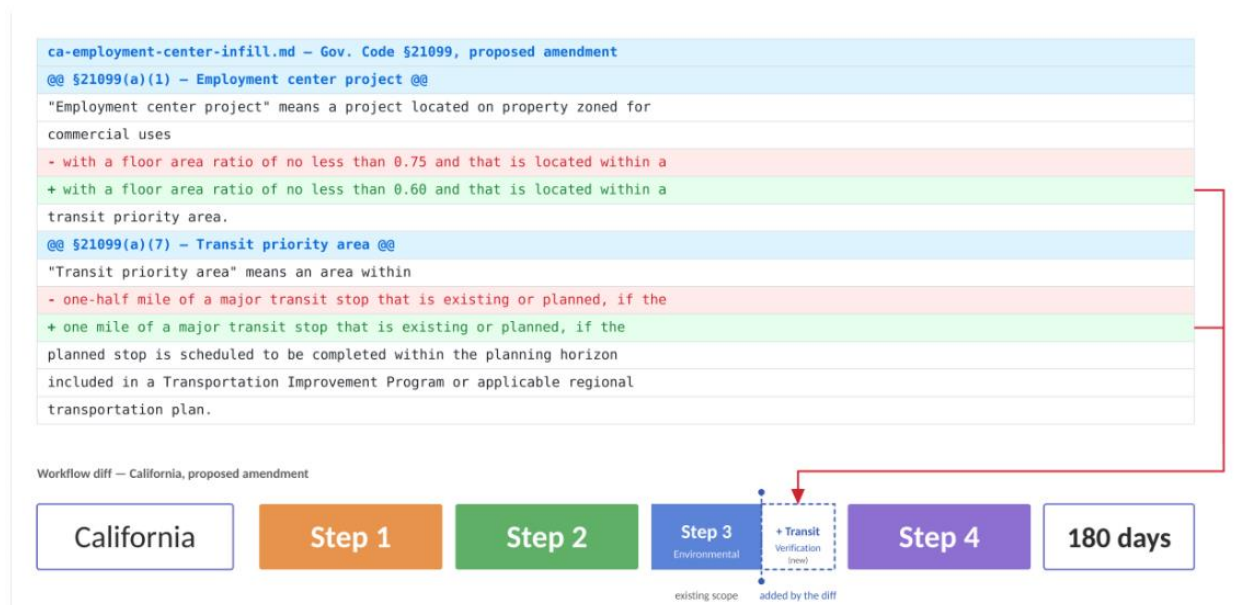


Figure 9 – Regulatory Modernization Packet “Dual Diff” Illustration: Once up and running, lawmakers can use ‘regulatory modernization packets’ proposed here to see both the difference in legal text side-by-side with the difference in workflow speed.

While a successful pilot does not automatically become law, pilot programs can be pre-certified to default to scaling or sunseting according to certified performance metrics. Under this setup, if performance metrics are achieved, lawmakers must explicitly intercede to stop a pilot from scaling – or in the case of the counterfactual – lawmakers must intercede to save a program if it does not meet performance metrics.

In any case, if the applicable legislative body (in this case, the Colorado legislature) approves, the packet containing the changes to the legal and operational schemas merges into the ‘production’ environment for the broader state. In short: the legislature authorizes the experiment, agencies implement, and independent certification bodies certify performance, which the legislature then approves.

6. Incorporating Automation for Long-Term Efficiency

Once the system is set up, the workflow portion of the schema may be connected via standard APIs to applicable agency software (be that permitting, inspection, procurement, appeals, and other applicable agency systems). In the long term, handoff between these agencies can become connected and timestamped events that automatically record telemetry mechanics as day-to-day operations proceed across a state.

The long-term goal is a government that has legibility into the differences between active agency work and queue time, locates bottlenecks and dependency failures, and so on – all the while lowering the documentation requirements that burden bureaucratic processes. Rather than exchanging information through sequential document transfers, participating agencies share operational and permitting telemetry through APIs with the workflow database. This gives lawmakers real-time feedback into the operations their work creates.



Figure 10 – Agency API Connections: Setting up inter-agency API data connections automates permitting telemetry and allows us to automate what would otherwise be inferred from siloed systems.

7. Version-Controlled Governance

This digital structure also enables lawmakers to search for broken dependencies, critical gaps in regulatory frameworks that have been solved elsewhere and enables lawmakers to ‘rollback’ changes if previously unidentified problems arise.

Preserving this structure also enables greater transparency for the public, as they can observe line-by-line authorship of laws by state or federal lawmakers, which also enables a greater degree of cross-examination if lawmakers write clauses and loopholes into thousand-page documents that suspiciously exempt their donors from taxes, regulatory changes, or similar.

8. Discussion

Governments have traditionally been great at creating or repealing laws, but they have persistently lacked the capability for parsing increasingly complex administrative systems those laws create.

This proposal is a technological maintenance tool designed for this very purpose. It connects the legal texts drafted by lawmakers to the operational workflows those texts produce. It makes

those workflows directly observable via dashboards, and provides governments with a disciplined pathway for piloting, evaluating, and improving regulatory frameworks over time.

If successful, this proposal would fundamentally improve the regulatory system from its current state. Legislatures and agencies would retain their authority to create laws and regulations but gain better tools and greater visibility into the operational workflows those regulations create.

With this proposal, lawmakers gain the ability to directly observe how regulations affect the construction timelines of solar farms, hospitals, houses, and other projects. They can compare their jurisdiction's regulatory environment with peer states to drive efficiency. They can easily identify agency bottlenecks and regulatory gaps. Finally, they can test changes to a regulatory framework before adopting them broadly.

The larger objective is to make regulatory systems legible as integrated software operations — currently known as DevOps. This enables continuous improvement and continuous delivery for government systems, along with a middle path between accumulated opaque regulations and indiscriminate repeal. Over time, this proposal shifts the equilibrium from a steady accumulation of regulatory layers towards a gradual adoption of the most efficient regulatory framework possible.

References

1. Fukuyama, Francis. *Political Order and Political Decay: From the Industrial Revolution to the Globalization of Democracy*. New York: Farrar, Straus and Giroux, 2014. pg. 514–517.
2. Teles, Steven M. *Kludgeocracy: The American Way of Policy*. Washington, D.C.: New America Foundation, 2012. pg. 5–6.
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5. Teles, S. M. (2013). "Kludgeocracy in America." *National Affairs*, 17, 97–114.
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